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United States Patent	12398327
Kind Code	B2
Date of Patent	August 26, 2025
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Apparatus and methods for the automatic cleaning and inspection systems of coke drums

Abstract

An integral system for automated and non-intrusive of cleaning and non-destructive inspection (ultrasonic volumetric testing and visual testing) to detect, characterize and monitor with precision the level of internal and external damage (Cracks, deformations, corrosion, erosion, etc.) that may be present in coke drums throughout their life cycle is disclosed. Embodiments are disclosed that enable a condition of a coke drum to be estimated in a reliable manner for their fitness for service from the results obtained from the automated inspection with the non-destructive methods of ultrasound, visual testing and/or liquid Penetrant Testing.

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Family ID: 1000008781983

Appl. No.: 18/355495

Filed: July 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240018415 A1	Jan. 18, 2024

Related U.S. Application Data

continuation-in-part parent-doc US 17447834 20210916 US 11835469 child-doc US 18355495

Publication Classification

Int. Cl.: C10B43/04 (20060101); C10B43/08 (20060101); C10B43/10 (20060101); G01N21/91 (20060101); G01N29/04 (20060101)

U.S. Cl.:

CPC C10B43/04 (20130101); C10B43/08 (20130101); C10B43/10 (20130101); G01N21/91 (20130101); G01N29/041 (20130101);

Field of Classification Search

CPC: C10B (43/04); C10B (43/08); C10B (43/10); C10B (33/006); C10B (45/00); C10B (55/00); G01N (21/91); G01N (29/041); G01N (21/954); B08B (9/093); B08B (9/0936)

USPC: 356/426

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3809607	12/1973	Murray et al.	N/A	N/A
4169758	12/1978	Blackstone et al.	N/A	N/A
4393711	12/1982	Lapides	N/A	N/A
4742713	12/1987	Abe et al.	N/A	N/A
5425279	12/1994	Clark et al.	N/A	N/A
5680215	12/1996	Huber et al.	N/A	N/A
5804730	12/1997	Pfannenstiel et al.	N/A	N/A
6104970	12/1999	Schmidt, Jr. et al.	N/A	N/A
7784347	12/2009	Messer et al.	N/A	N/A
7940298	12/2010	Clark et al.	N/A	N/A
8436898	12/2012	Clark et al.	N/A	N/A
9524542	12/2015	Clark et al.	N/A	N/A
2005/0151841	12/2004	Nelson et al.	N/A	N/A
2006/0288756	12/2005	De Meurechy	N/A	N/A
2007/0000328	12/2006	Buttram	N/A	N/A
2010/0007729	12/2009	Clark et al.	N/A	N/A
2011/0272617	12/2010	Lah	N/A	N/A
2013/0329848	12/2012	Linnebur	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103558293	12/2013	CN	N/A
2003013070	12/2002	JP	N/A
2011237234	12/2010	JP	N/A

Primary Examiner: Rahman; Md M

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/079,139 filed 16 Sep. 2020 and is a continuation-in-part of co-pending U.S. patent application Ser. No. 17/447,834 filed 16 Sep. 2021. The disclosure of the applications above are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) Embodiments of the present disclosure relate to an automated and non-intrusive integral system and method of cleaning and inspection by non-destructive testing; visual testing (VT) and ultrasonic testing (UT) capable of detecting and characterizing internal and external cracks regardless of the nominal thickness of the cylindrical, cone-skirt and conical sections in delayed coke drums.

(2) In certain embodiments, the results obtained by the ultrasonic inspection, it will be possible to estimate the condition of the coke drums by their fitness for service and in this manner to increase the operational reliability of these pressure vessels.

DESCRIPTION OF THE RELATED ART

(3) Around the world, the oil industry uses coke drums as a technology for the upgrade of heavy oil and coke production. This process is affected by the decrease in the life of coke drums caused by the phenomenon of low cycle fatigue, generating cracks and deformations that must be inspected and repaired, affecting the continuous operation of the delayed coking units.

(4) Most coke drums are constructed with a low alloy steel specification ASTM A-387 Gr 11 ($1\frac{1}{4}\text{Cr}-\frac{1}{2}\text{Mo}$), as well as ASTM A-387 Gr 12 ($1\text{Cr}-\frac{1}{2}\text{Mo}$) or ASTM A-387 Gr 22 ($2\frac{1}{4}\text{Cr}-1\text{Mo}$). The thicknesses can vary depending on the elevation and location of the plates of these pressure vessels, which become thinner from the bottom to the top; they are usually from 1 in to 2 in of Thickness (25.4 mm to 50.8 mm). They can have a cladding of ASTM A-240 type 410 with a thickness of 3.2 mm ($\frac{1}{8}$ in), which gives them a high resistance to corrosion and abrasion. The horizontal or circumferential welds that attach to each of the plates can have an inner and outer crown approximately 50 cm (19.68 in) of wide and are flat flush with the surface.

(5) Currently, operators (Refineries and Upgraders) of coke drums worldwide are decreasing the time of operating cycles in a half, which will result in the appearance of cracks, bulges or other defects in a shorter time in these pressure vessels. As a result, it is necessary that the interior and exterior of such pressure vessels be frequently cleaned and inspected using non-destructive testing techniques. In the past and at present, such volumetric inspections (i.e. detect defects in the entire volume of the material as opposed to just surface defects) have been achieved in the following ways: 1. Emptying the vessel, installing internal scaffolds and manually and physically inspecting the interior surfaces of the vessel, detecting deformations, cracks and other defects; 2. Installing external scaffolds, disassembling the thermal insulation and inspect manually from the outer surface of the vessel to detect deformations, cracks and other defects; and 3. Rappelling by manually inspecting from the outer or interior surface of the container to detect deformations, cracks and other defects.

All of the above are performed in an intrusive manner.

(6) At present, there are no automated technologies to estimate the condition of coke drums and to evaluate their fitness for service in a reliable manner through the results of a non-intrusive volumetric inspection, regardless of the origin of the crack (internal, external or embedded in the cladding) and the thickness of the vessel wall.

(7) For each of these non-destructive inspections (volumetric as and no volumetric) in a coke drum, or in any pressure vessel, the inspections must be carried out after an adequate cleaning of the welds or areas of interest in order to obtain a surface finish that allows obtaining reliable results.

(8) At present, if it is required to perform an internal cleaning of a coke drum in a non-intrusive manner, this is only done through the use of a cutting tool and it does not tend to be effective. In addition, there is no automated inspection system and method in the world able to detect and

characterize the cracks in the AID and AOD areas as defined by American Petroleum Institute API 1996 (Coke Drum Survey) and where the "A" corresponds identifies a crack in the weld of the shell or body of the container and the acronym ID and OD indicate whether the crack is located in the inner or outer diameter of the cone-skirt joint of a coke drum.

(9) The systems, methods and technologies of non-intrusive and intrusive inspection in coke drums of the prior art have at least the following limitations: I. Highly expensive; II. When the inspection is internal it requires the use of scaffolding III. When the inspection is external it requires the use of scaffolding and the removal of thermal insulation IV. They do not perform efficient cleaning to obtain optimal surface finishes that guarantee reliable results; V. Manual methods require long times to clean and inspect; VI. In manual methods, personnel are exposed to possible accidents (falls, gases, etc.); VII. Automated methods do not detect external cracks; VIII. The automated methods detect internal cracks if and only if they are connected to the internal surface of the pressure vessel; IX. In automated methods, if the cracks are in the interface or fusion zone between the base metal and the overlay, they do not detect them; X. Automated methods have limitations to measure depth of cracks; XI. Automated methods cannot detect cracks in an AID zone located in the cone-skirt joint of the coke drums; and XII. Automated methods do not allow their use as reliable input for the fitness for service of cracked internal and external areas in the sections: cylindrical, cone-skirt junction and conical for coke drums.

(10) What is needed is improved intrusive and non-intrusive inspection methodologies and systems for the internal and external cleaning and volumetric inspection of coke drums

SUMMARY OF THE INVENTION

(11) A system of one or more computers can be configured to perform particular operations or actions by virtue of having software, firmware, hardware, or a combination of them installed on the system that in operation causes or cause the system to perform the actions. One or more computer programs can be configured to perform particular operations or actions by virtue of including instructions that, when executed by data processing apparatus, cause the apparatus to perform the actions. One general aspect includes an adjustable ramp system configured to transfer an inspection crawler into and out of an interior space of the coke drum the adjustable ramp may include a horizontal section having a horizontal length which may include a drive gear rotatably positioned on a first end, a first guided gear in meshing arrangement with the drive gear coupled to a first worm screw rotatably mounted to the horizontal section, a second guided gear in meshing arrangement with the drive gear coupled to a second worm screw rotatably mounted to the horizontal section a vertical section having an adjustable vertical length, a first vertical drive shaft threadably engaged within a portion of the horizontal section and in meshing arrangement with the first worm screw, a second vertical drive shaft threadably engaged within a portion of the horizontal section and in meshing arrangement with the second worm screw an angular displacement device positioned on a bottom end of the vertical section configured to be angularly adjustable relative to the vertical section and a front section having an adjustable front section length coupled to the angular displacement device and a ramp surface having an adjustable ramp length positioned on a portion of the horizontal section, the vertical section and the front section configured to provide a contiguous surface. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(12) Implementations may include one or more of the following features. The coke drum inspection system where the horizontal section is configured to couple to an inlet of the coke drum. The adjustable vertical length is configured to be adjusted by rotation of the drive gear to accommodate a length of the inlet of the coke drum. The inlet of the coke drum may include any of a flange and a sliding valve. The angular displacement device is configured to angularly adjust the front section to a predetermined angle relative the vertical section and the front section is configured to adjust the front section length to contact an inside surface of the interior space of the coke drum at a

predetermined location. The inspection crawler may include a chassis and a drive system mounted to the chassis, a reception device configured to allow a remote control operation of the inspection crawler, and at least one non-destructive testing device mounted to the inspection crawler. At least a portion of the drive system may include a magnetic material, the ramp surface may include a ferrous material and the coke drum may include a ferrous material and where the drive system is configured to propel the inspection crawler along the ramp surface and along the inside surface of the interior space of the coke drum. The at least one inspection device may include an ultrasonic testing system. The ultrasonic testing system may include an articulating frame mounted to the chassis, and a plurality of ultrasonic testing probes mounted to the frame and configured to contact a portion of the inside surface of the interior space of the coke drum. The at least one inspection device may include a visual testing system. The visual testing system may include a camera and a lighting system. The coke drum inspection system may include a lighting system positioned on the front section. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

(13) One general aspect includes a method of inspecting a coke drum. The method of inspecting also includes providing an adjustable ramp system having a horizontal a vertical section having an adjustable vertical length, an angular displacement device positioned on a bottom end of the vertical section and a front section having an adjustable front section length coupled to the angular displacement device and a ramp surface having an adjustable ramp length positioned on a portion of the horizontal section, the vertical section and the front section configured to provide a contiguous surface. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(14) Implementations may include one or more of the following features. The method of inspecting a coke drum may include coupling the horizontal section to an inlet of the coke drum. The method of inspecting a coke drum may include adjusting the vertical length to accommodate a length of the inlet of the coke drum. The method of inspecting a coke drum may include adjusting the angular displacement device, and adjusting the front section length and contacting an inside surface of an interior space of the coke drum at a first predetermined location. The method of inspecting a coke drum may include providing an inspection crawler having a chassis and a drive system mounted to the chassis and at least one non-destructive testing device mounted to the inspection crawler, placing the inspection crawler on the adjustable ramp, and remotely controlling the inspection crawler from outside of the interior space of the coke drum. The method of inspecting a coke drum may include propelling the inspection crawler along the ramp surface and along the inside surface of the interior space of the coke drum. The method of inspecting a coke drum may include inspecting the coke drum using the at least one non-destructive testing device. The method of inspecting a coke drum may include controlling the inspection crawler to the adjustable ramp and removing the inspection crawler from the coke drum. The method of inspecting a coke drum may include adjusting the angular displacement device, adjusting the front section length and contacting an inside surface of the interior space of the coke drum at a second predetermined location, placing the inspection crawler on the adjustable ramp, remotely controlling the inspection crawler from outside of the interior space of the coke drum, propelling the inspection crawler along the ramp surface and along the inside surface of the interior space of the coke drum, and inspecting the coke drum using the at least one non-destructive testing device. The at least one inspection device may include an ultrasonic testing system, the method may include ultrasonically inspecting the coke drum. The ultrasonic testing system further may include a plurality of ultrasonic testing probes, and positioning the plurality of ultrasonic testing probes to contact the inside surface of the interior space of the coke drum. The at least one inspection device may include a visual testing system, the method may include visually inspecting the interior space of the coke drum. The method of inspecting a coke drum may include providing a lighting system positioned on the front section and

lighting the interior space of the coke drum. Implementations of the described techniques may include hardware, a method or process, or computer software on a computer-accessible medium.

(15) In one general aspect, a coke drum cleaning system may include a vertical member. The coke drum cleaning system may also include a gear box fixedly attached to a distal end of the vertical member including a joint. The system may furthermore include a gear box motor coupled to the gear box. The system may in addition include an arm pivotally attached at a proximal end to the joint and configured to pivot from a vertical position to a horizontal position. The system may moreover include a sliding member selectively slidably positioned within the arm. The system may also include a cleaning mechanism mounted to a distal end of the sliding member. The system may furthermore include where the gear box motor is configured to selectively rotate the arm about the vertical member in the horizontal position.

(16) Implementations may include one or more of the following features. The coke drum cleaning system may include a retractable mechanism motor coupled to the sliding member, and where the retractable mechanism motor is configured to selectively position the sliding member between a retracted position and a plurality of extended positions. The coke drum cleaning system may include an approach sensor configured to sense a wall of a coke drum and is further configured to position the cleaning mechanism proximate a wall of the coke drum with the arm in the horizontal position. The coke drum cleaning system may include a cleaning pipe system coupled to the vertical member and in fluid communication with the cleaning mechanism. The coke drum cleaning system may include a plurality of nozzles positioned on the cleaning mechanism in fluid communication with the cleaning pipe system. The coke drum cleaning system may include a metal cord coupled to the arm and to the gear box, a ring slidably mounted to the vertical member where the metal cord is slidably positioned within the ring, and where the metal cord is configured to be slid vertically to pivot the arm from a vertical position to a horizontal position. The coke drum cleaning system where the coke drum includes an upper nozzle positioned in a top of the coke drum and where coke drum cleaning system is configured to be introduced into an interior of the coke drum through the upper nozzle with the arm in the vertical position. The coke drum cleaning system where the arm is positioned in a cleaning position with the arm in the horizontal position.

(17) In one general aspect, system may include a vertical member. The system may also include a cutting tool attached to a lower end of the vertical member. The system may furthermore include a plurality of cleaning nozzles mounted to the cutting tool. Other embodiments of this aspect include corresponding computer systems, apparatus, and computer programs recorded on one or more computer storage devices, each configured to perform the actions of the methods.

(18) Implementations may include one or more of the following features. The system may include a cleaning pipe system coupled to the vertical member and in fluid communication with the plurality of cleaning nozzles. The system where a coke drum includes an upper nozzle positioned in a top of the coke drum and where the cutting tool is configured to be introduced into an interior of the coke drum through the upper nozzle using the vertical member. The system where the vertical member is configured to rotate within the interior of the coke drum during a cleaning operation.

(19) In one general aspect, method may include providing a coke drum cleaning system having a vertical member, a gear box fixedly attached to a distal end of the vertical member including a joint, a gear box motor coupled to the gear box, an arm pivotally attached at a proximal end to the joint, a sliding member selectively slidably positioned within the arm, and a cleaning mechanism mounted to a distal end of the sliding member. Method may also include introducing the coke drum cleaning system into an interior of the coke drum. Method may furthermore include pivoting the arm from a vertical position to a horizontal position. Method may in addition include rotating the arm using the gear box motor about the vertical member in the horizontal position.

(20) Implementations may include one or more of the following features. Method may include providing a retractable mechanism motor coupled to the sliding member, and positioning the sliding member between a retracted position and a plurality of extended positions using the

retractable mechanism motor. Method may include an approach sensor mounted to the arm, sensing a wall of a coke drum using the approach sensor, and positioning the cleaning mechanism proximate the wall of the coke drum with the arm in the horizontal position. Method may include coupling a cleaning pipe system to the vertical member and in fluid communication with the cleaning mechanism. Method may include coupling a plurality of nozzles positioned on the cleaning mechanism in fluid communication with the cleaning pipe system. Method may include a metal cord coupled to the arm and to the gear box, a ring slidably mounted to the vertical member where the metal cord is slidably positioned within the ring, sliding the metal cord vertically, and pivoting the arm from a vertical position to a horizontal position. Method having any of water blasting, sandblasting, shot peening, laser cleaning, and material impacting on an inner surface of the coke drum. Method may include providing a coke drum cutting tool, replacing a plurality of cutting nozzles in the coke drum cutting tool with a plurality of cleaning nozzles, and cleaning an inner surface of the coke drum using the plurality of cleaning nozzles. Method may include preparing an inner surface of the coke drum for non-destructive inspection. Method may include performing liquid penetrant testing using the coke drum cleaning system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) So that the manner in which the above-recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, can be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.
- (2) FIG. 1A is a detailed section view of an arm or support structure of a cleaning mechanism inside a delayed coke drum in accordance with the present disclosure;
- (3) FIG. 1B is a detailed section view of an arm support structure of a cleaning mechanism inside a delayed coke drum in accordance with the present disclosure;
- (4) FIG. 2 is a detailed section view of an arm support structure of a cleaning mechanism articulated at 90° with respect to the cutting tool inside a delayed coke drum in accordance with the present disclosure;
- (5) FIG. 3 shows inside a delayed coke drum the support structure of the cleaning mechanism articulated at 90° with respect to the cutting tool and extended towards the internal wall of the pressure vessel in accordance with the present disclosure;
- (6) FIG. 4 shows details of end of the cleaning device near to the internal wall of the pressure vessel in accordance with the present disclosure;
- (7) FIG. 5A is an isometric view of a UT and VT inspection crawler for coke drums in accordance with the present disclosure;
- (8) FIG. 5B is a top view of a UT and VT inspection crawler for coke drums in accordance with the present disclosure;
- (9) FIG. 5C is a left side view of a UT and VT inspection crawler for coke drums in accordance with the present disclosure;
- (10) FIG. 5D is rear view of a UT and VT inspection crawler for coke drums in accordance with the present disclosure;
- (11) FIG. 6A is a detailed view of a coupling of an adjustable ramp to the upper flange of the delayed coke drum for positioning an UT and VT inspection crawler on its internal surface in accordance with the present disclosure;
- (12) FIG. 6B is a detailed view of a coupling of an adjustable ramp to the upper flange of the

delayed coke drum for positioning an UT and VT inspection crawler on its internal surface in accordance with the present disclosure;

(13) FIG. 6C is a detailed view of a coupling of an adjustable ramp to the upper flange of the delayed coke drum for positioning an UT and VT inspection crawler on its internal surface in accordance with the present disclosure;

(14) FIG. 6D is a detailed view of a coupling of an adjustable ramp to the upper flange of the delayed coke drum for positioning an UT and VT inspection crawler on its internal surface in accordance with the present disclosure;

(15) FIG. 7A is an isometric view of an adjustable ramp for the entry, positioning and extraction of the UT and VT inspection crawler from the interior of the coke drum in accordance with the present disclosure;

(16) FIG. 7B is top view of an adjustable ramp for the entry, positioning and extraction of the UT and VT inspection crawler from the interior of the coke drum in accordance with the present disclosure;

(17) FIG. 7C is left side view of an adjustable ramp for the entry, positioning and extraction of the UT and VT inspection crawler from the interior of the coke drum in accordance with the present disclosure;

(18) FIG. 7D is front view of an adjustable ramp for the entry, positioning and extraction of the UT and VT inspection crawler from the interior of the coke drum in accordance with the present disclosure;

(19) FIG. 8 is a detailed view of the displacement of the inspection crawler of UT and VT through the adjustable ramp into the delayed coke drum in accordance with the present disclosure;

(20) FIG. 9 is a detailed view of the displacement of the UT and VT Crawler towards the exterior of the delayed coke drum by use of the ramp in accordance with the present disclosure;

(21) FIG. 10A is a detailed view of a cone-skirt junction of a coke drum showing the positioning of the crawler for the detection and characterization of cracks in the susceptible zones of these cracks (AID and AOD areas) in accordance with the present disclosure;

(22) FIG. 10B is a detailed view of a cone-skirt junction of a coke drum showing the positioning of the crawler for the detection and characterization of cracks in the susceptible zones of these cracks (AID and AOD areas) in accordance with the present disclosure;

(23) FIG. 11 is a flow chart of a method of inspecting a coke drum in accordance with the present disclosure;

(24) FIG. 12 is a side view of a coke drum cleaning system in accordance with the present disclosure; and

(25) FIG. 13 is a flow chart of a method of cleaning a coke drum in accordance with the present disclosure.

DETAILED DESCRIPTION

(26) In the following detailed description of the embodiments, reference is made to the accompanying drawings, which form a part hereof, and within which are shown by way of illustration specific embodiments by which the examples described herein can be practiced. It is to be understood that other embodiments can be utilized, and structural changes can be made without departing from the scope of the disclosure.

(27) With reference to FIGS. 1A and 1B, the method of embodiments of the present disclosure is carried out in such a way that cleaning and inspection system 23 is supported on cutting tool stem 25 and it is introduced through upper flange 17 of delayed coke drum 15, The inspection system 23 is comprised of a reducer gear box 50 attached to cutting tool stem 25 using a clamp, this gear box 50 is activated by electric motor 40a that induces rotational movement to the inspection system around the cutting tool. Referring next to FIGS. 2 and 3, reducer gear box 50 is connected to an arm or support structure by an articulated joint 24 containing a cleaning mechanism 26 including a pipe system 280, in order to perform the cleaning of the internal surface 16 of the welds of the

pressure vessel or any area of coke drum **15** that it is desired to clean for an inspection (FIG. **3**). It should be appreciated by those skilled in the art that non-destructive inspection requires a good cleaning of the coupling surface to obtain satisfactory results. Support structure **23** is articulated manually by a metal cord **260** which is attached to the support structure **23** through a ring **170** attached to the support structure and a ring **160** connecting the metal cord to reducer gear box **50**. (28) Still referring to FIG. **3**, video device **21** comprised of a video camera is supported in the bottom section of the cleaning system **23** and is protected within a gas capsule provided by a system pipe **290**, this video camera captures videos and images of circumferential, longitudinal welds and any part of the inner surface **16** of the pressure vessel during cleaning time. Video device **21** is capable of moving its visual field and focus at different angles and can be assisted by a lighting mechanism that provides a better definition of the images. Visual data is collected by video device **21** and the data collected is stored in a computer that is located on the outside of the pressure vessel **15**. Inside the gas capsule and positioned next to the video camera is also an approximation laser sensor, capable of measuring the distance with the area focused by the video device **21**. Referring to FIG. **4**, there is shown cleaning device **26** located at the end of the structure of the cleaning mechanism **23** wherein the cleaning device is configured to work with water and/or solid particles under pressure and this manner ensure the necessary cleaning for inspection. The cleaning device **26** is comprised of pipe system **280** (FIG. **3**) containing a length adapter **27** including a plurality of holes and a holder introduced inside the selected hole of the length adapter. Additionally, a retractable mechanism is also activated through motor **40b** (FIG. **3**) which is capable to bring the plurality of cleaning nozzles of cleaning device **26** closer to the internal surface of the coke drum **16** through a sliding system **77** by means of an approach sensor **60** in order to reach the optimum distance for the cleaning operation. Once the internal surface **16** has been visually inspected and cleaned, reaching the appropriate surface finish, the UT inspection as will be disclosed in more detail herein below:

(29) With respect to FIG. **1B**, in embodiments of delayed coke drums **15** having sliding valves in upper section **100**, it should be appreciated by those skilled in the art that it can be necessary to perform extra maneuvers to manipulate for cutting tool stem **25** carefully as well as the entry and exit from pressure vessel **15** in order to avoid damaging any of the internal components of the sliding valves.

(30) Generally referring to FIGS. **6A-6D**, after having carried out the remote visual inspection and the cleaning of the areas of interest for the UT inspection as disclosed herein above, the cleaning and inspection system **23** (FIGS. **1A, 1B**) can be extracted from the interior **51** of the coke drum **15** and wherein an adjustable ramp system comprises an adjustable access ramp **14** having an adjustable ramp length can be fitted which can be used for accessing the upper flange **17** (FIGS. **6A, 6B**) as well as the lower flange **28** (FIGS. **6A, 6C**) of the delayed coke drum **15** wherein the front section of the ramp can be contacted at a predetermined location of the coke drum. If the coke drum **15** to be inspected has slide valves (FIGS. **6B, 6C**), in the case of the stop valve **100** the adjustable ramp **14** will be supported in the Stem Guide Enclosure Cover Sheet **95** (FIG. **6B**). It should be appreciated by those skilled in the art that adjustable access ramp **14** is configured to transfer inspection crawler **14** into and out of the interior of coke drum **15**. The ramp surface of adjustable access ramp **14** is configured such that as the adjustable ramp length is changed that there exists a contiguous surface from one end of the ramp to the other. In embodiments including a lower slide valve **200**, adjustable ramp **14** will be supported in the lower section of the valve. With specific reference to FIGS. **8** and **9**, adjustable ramp **14** is shown positioned in the lower section of lower slide valve **200** in order to allow the entry, positioning and extraction of the UT and VT inspection crawler **1** (FIGS. **5A-5D**) in the inner surface of the pressure vessel **16**. UT and VT inspection crawler **1** includes an ultrasonic testing system and a visual testing system and is operated from the outside of the coke drum **15** by remote control and the operator performs the crawler movement by means of the video camera **6** (FIGS. **5A-5D**) positioned at the rear part of UT

and VT inspection crawler **1**.

(31) The electrical parts of the devices used (cameras, UT equipment, etc.) in the cleaning and inspection system and the inspection crawler **1** are encapsulated and protected by a gas pressure higher than the internal pressure of the coke drum **15**.

(32) The UT and VT signals obtained from the UT reception and emission device of the cylindrical sections **16** of coke drum **15**, conical section **18** (FIG. **8**) and cone-skirt junction of cone **300** (FIG. **10A**) and skirt **310** (FIG. **101B**) are transmitted to a computer located on the outside of the coke drum **15** through a wiring system **8** (FIG. **5A-5D**) specific for this purpose, at the same time, the UT probes **3** located in the front of inspection crawler **1** have an input for the irrigation of the coupling between the UT probes and the coke drum **15** and this is supplied by a system of hoses **9** coming from the outside of the coke drum.

(33) Once the UT inspection is completed, UT and VT inspection crawler **1** is removed from the inside **51** of the coke drum **15** by use of the adjustable ramp **14** (FIG. **9**) located in the upper nozzle **17** or lower nozzle **28** of the drum and finally it removes the access ramp **14**, leaving the pressure vessel **15** available for operation.

(34) FIGS. **1A** and **1B**, the support structure of the cleaning mechanism inside a delayed coke drum is shown which can even work with or without sliding valves.

(35) Referring to FIGS. **1A** and **1B**, the reducer gear box **50** is coupled to the cutting tool stem **25** of the coke drum **15** and activated in its rotational movement by a motor **40a** and is remotely maneuvered from the outside of the pressure vessel. FIGS. **1A** and **1B** differ in that the entry maneuvers of the cleaning and inspection system **23** can be executed in delayed coke drums **15** with or without sliding valves.

(36) As shown in FIGS. **1A**, **1B**, a typical delayed coke drum **15**, has two closing flanges that guarantee the tightness of the pressure vessel and that also serve for entry of personnel to perform cleaning and intrusive inspections, as well as the entrance of the cutting tool stem **25**. The upper flange **17** and the lower flange **28** are of diameters greater than 24 inches and in both of them sliding valves **100** and **200** can be respectively connected. The wall thickness of the drum **15** is greater than 1 inch and may vary with its elevation. In addition, the external surface of the pressure vessel **16** is covered or thermally insulated (not shown).

(37) Referring to FIGS. **1A**, **1B**, cutting tool stem **25** is generally comprised of a cylindrical pipe and a drill (not shown), which are introduced from the upper part of the drum **15**, by the upper flange **17** moving vertically downwardly through the internal part of the pressure vessel **51** to the lower flange **28** where the product "Coke" is downloaded. Cutting tool stem **25** has the function of cutting the coke inside **51** of coke drum **15**. In embodiments of the present disclosure, cutting tool stem **25** is equipped with cleaning and inspection system **23** and is configured to introduce the cleaning and inspection system inside the pressure vessel **51** and to move it vertically therein. The movement of cutting tool stem **25** is controlled by an independent motor of the cleaning and inspection system **23** and at the time of cleaning and inspection, the cutting tool is held stable and adjusted in embodiments having a superior sliding valve. In other embodiments Stem Guide Enclosure Cover Sheet **95** (FIG. **6B**) provides the stability and adjustment of the cutting tool. Turning motor **40a** is configured to control the radial movement of inspection system **23**, wherein the turning motor independently controls the circumferential movement of the inspection system with respect to the cutting tool stem **25**.

(38) Referring again to FIGS. **2**, **3**, support structure **23** contains a remote visual inspection device (RVI) that is comprised of a video and photography camera **21** with built-in lighting, which are inside a capsule that manages a gas or air current provided by a hose **290** that serves as protection for the electronic components. An approximation laser device **22** is positioned in the central axis of the remote visual inspection system RVI wherein the approximation laser device capable of measuring the distance between the front of the camera **21** and the internal wall of pressure vessel **16**.

(39) The video and photo camera **21** contained in the capsule is capable of being moved in both the radial direction (circumferentially) and vertical direction (up and down), which enables the camera to be tilted before focusing on any particular area within pressure vessel **16**. Video and photography camera **21** is preferable of high resolution capable of being used by those skilled in the art.

(40) The video and photography camera **21** contained in the capsule beyond having a first lighting control, is assisted by a second complementary lighting system in which each source can act independently and the intensity of each of the sources can be adjusted.

(41) As disclosed herein above, cleaning system **23** enters interior **51** of coke drum **15** downwardly in the vertical direction coupled to the cutting tool stem **25** through the upper nozzle **17**, or through the Stem Guide Enclosure Cover Sheet **95** (FIG. 6B). Once cleaning system **23** is positioned inside the coke drum **15**, it is actuated by articulation **24** located at an end of the gear box **50** allowing the support structure of the cleaning and inspection system **23** (FIG. 3) to be positioned 90 degrees to cutting tool stem **25**. The cleaning portion of cleaning and inspection structure **23** is comprised of hose **280** and a manifold of nozzles **26** (FIG. 4) which has the ability to move horizontally with a retractable mechanism with the use of a motor **40b** (see FIG. 3). The rotational movement in circumferential direction of cleaning and inspection system **23** is controlled by remote control independently of the cutting tool stem **25** by using the motor **40a**, in order to perform the cleaning of the areas to be inspected with UT and VT inspection crawler **1**.

(42) Referring now to FIGS. 5A-5D, there is shown the detail of UT and VT inspection crawler **1**. The UT and VT inspection crawler **1** includes a laser that has the objective of guiding it through the welded joint to be inspected considering an equidistant position between UT probes **3**. Once UT and VT inspection crawler **1** is positioned on the inner surface **16** of the coke drum **15**, UT probes **3** are remotely adjusted within an articulating frame mounted to the chassis of the inspection crawler by means of the articulate junction **33** to the distances established by predetermined scan plans prepared for the inspection procedures by UT. It should be appreciated by those skilled in the art that such pans can be comprised of Phased array and Time-of-flight diffraction (TOFD) time testing and other known ultrasonic techniques. UT probes **3** have an irrigation system that supplies a fluid to couple probes to the wall of coke drum **15** to guarantee the transmission, emission and reception of UT signals. The irrigation system of UT probes **3** is fed by hose **9** that is coupled to a fluid supply positioned on the outside of the coke drum **15**. The inspection crawler of UT and VT **1** is operated by remote control and its movement on the internal surface **16** of the coke drum **15** can be accomplished using a drive system such as with the use of magnetic wheels **4a**, **4** as well as a caterpillar system (not shown) made of a magnetic material. In the case of using magnetic wheels, inspection crawler of UT and VT **1** includes four magnetic wheels **4** of which one also functions as encoder **4A** for the registration of its displacement. The interior of coke drum **15** and the ramp surface of adjustable access ramp **14** can comprise a ferrous material to cooperate with the magnetic wheels **4**, **4a** to keep the inspection crawler adhered to their surfaces. In the upper section of inspection crawler of UT and VT **1** is emission and reception device **31**, that produce emission and reception signals carried by a wiring **8** to a computer on the outside of the coke drum **15** where they are analyzed and stored. On the back of UT and VT inspection crawler **1** at a certain height is a lighting system comprised of two lamps **5** located on each side of a video camera **6** with a visual field **7** that will enable a user to maneuver of the inspection crawler on the internal surface **16** of the coke drum **15** and for visual inspection of the areas to be analyzed inside the coke drum **51**.

(43) A method for cleaning and inspecting coke drum **15** is characterized by the introduction of cleaning and inspection system **23** into the coke drum, in order to perform the visual inspection and cleaning of the areas that require it to adapt them to the surface finish required for the inspection.

UT with UT and VT inspection crawler **1**, taking into account the procedures mentioned below:

(44) Shown in FIGS. 1A, 1B are the details of the introduction of cleaning and inspection system **23** into the interior **51** of the coke drum **15**, in which it is observed an entry of the cleaning and inspection system **23** in the vertical downward direction using access through the upper nozzle **17**

or through the Stem Guide Enclosure Cover Sheet **95** (FIG. **6B**) of the coke drum **15**.

(45) Shown in FIG. **2** is the 90° articulation (by gravity in some embodiments) of the support structure of the cleaning and inspection system **23** inside **51** of the coke drum **15**.

(46) Shown in FIGS. **3**, **4** is the deployment of the cleaning manifold **26** by means of a retractable mechanism towards the internal wall **16** of the coke drum **15** and it is controlled by the approach sensor **60** in order to guarantee that there is the optimum distance to perform an adequate surface cleaning.

(47) Once the remote visual inspection and the cleaning of the identified zones to be inspected with UT have been carried out, the inspection and cleaning system **23** of the coke drum **15** is removed by reversing the steps immediately disclosed herein above.

(48) FIGS. **6A**, **6B**, **6C** and **6D** show adjustable access ramp **14** as being configured to couple to, and positioned in, an inlet to coke drum **15**, such as the upper nozzle **17** (see FIG. **6A**), in the Stem Guide Enclosure Cover Sheet **95** (FIG. **6B**), in the lower section of the valve slide **200** (see FIG. **6C**) or in the bottom flange **28** (see FIG. **6D**) of the coke drum. Adjustable access ramp **14** includes a horizontal section having a horizontal length and a vertical section having an adjustable vertical length. Adjustable access ramp **14** has the purpose of allowing the entry and extraction of the UT and VT inspection crawler **1** into and out of an interior space **51** of coke drum **15** and travels along and inside surface of the interior space. Referring next to FIGS. **7A-7D**, adjustable access ramp **14** is characterized by having a set of mechanism gears between which include drive gear **11** and two guided gears **10** in a meshing arrangement that respectively drive the rotation of two horizontal worm screws **12** which respectively drive vertical drive shafts **13**. Vertical drive shafts **13** have a thread on their surface that generate an upward and downward movement of the vertical section of the ramp adjusted by rotation of the drive gear in order to have an adjustable vertical length to adjust the height of the upper nozzle **17**, or considering the corresponding heights of the sliding valves either in their upper section **100** or lower **200** (FIG. **1B**). Adjustable access ramp **14**, in addition to being adjusted horizontally and vertically, is also angularly adjustable, by means of an angular displacement device **90** mounted to a bottom end of the vertical section and comprised of a semi-circumferential surface that allows an angular displacement relative to the lower section of the adjustable access ramp to a predetermined angle. Adjustable access ramp **14** includes lighting system **80** located on its lower sides and front section having an adjustable front section length that serves as a guide for the extraction of UT and VT inspection crawler **1**.

(49) It should be appreciated by those skilled in the art that FIGS. **6A**, **6B**, **6C** and **6D** disclose the procedure of introducing UT and VT inspection crawler **1** to the inside **51** of the coke drum **15** with the use of the adjustable access ramp **14**. In addition, FIGS. **8** and **9** show the positioning and removal of UT and VT inspection crawler **1** from the interior **51** of the coke drum **15** with the use of the adjustable ramp **14**. In this particular embodiment the procedure is shown in a coke drum **15** without sliding valves **100** and **200**. It should be noted that the UT and VT inspection crawler **1** introduction into and removal maneuver from the inside **51** is similarly performed in coke drums **15** with sliding valves **100** and **200** (FIG. **1B**). Once the inspection and removal of the crawler from inside **51** of the coke drum **15** is completed, a user can remove the access ramp **14** from the upper nozzle **17** or from the respective sliding valves **100** and **200**.

(50) Embodiments of the present disclosure allow a totally independent operation of UT and VT inspection crawler **1** with respect to the cleaning and inspection structure **23**, this characteristic is here to fore unknown in the automated systems of inspection of coke drums. Another advantage of embodiments of the present disclosure over the prior art is that UT and VT inspection crawler **1** can detect and characterize internal and external cracks regardless of the thickness of the wall in the cylindrical sections **16**, conical section **18** and cone-skirt joint section. In this last section for the detection and characterization of the cracks AID **300** FIG. **10A** and AOD **310** FIG. **10B** of the delayed coke drums.

(51) Now with reference to FIG. **11**, there is shown a flow chart of a method of inspecting a coke

drum in accordance with the present disclosure.

(52) Now with reference to FIG. 12, there is shown cleaning tool **34** comprised of modified cutting tool **36** coupled to flange **35** which is coupled to cutting tool stem **25**. Modified cutting tool **36** comprises a cutting tool of the prior art with cleaning nozzles **37**, **38** installed to replace the cutting nozzles. In addition to the cleaning methods disclosed herein above, a first generalized cleaning can be performed using cleaning tool **34**. In operation, cleaning tool **34** is lowered into interior space **51** of coke drum **16** (FIG. 1A). As cleaning tool **34** traverses the interior space **51** cleaning solutions are pumped through cutting tool stem **25** at a high pressure where they exit through cleaning nozzles **37** and impinge upon the walls of coke drum **16**. Such cleaning solutions can include water blasting, sandblasting, shot peening or laser cleaning, material impacting and the like. Cutting stem **25** can be rotated to position cleaning nozzles **37**, **38** at various radial positions within interior space **51** and can be rotated continuously while cleaning tool **34** traverses the length of the interior space to insure proper cleaning of the entire interior space. During the cleaning operation the cleaning solution can exit the interior space **51** through nozzle **28** (FIG. 1A) and can be captured for further use or proper disposal. This general cleaning method can be performed multiple times and can be performed after the cleaning methods disclosed herein above without departing from the scope of the present disclosure.

(53) Referring back to various figures, an integral system and method of cleaning and non-intrusive volumetric inspection of delayed coke drums is described to inspect the internal and external surfaces in order to detect, characterize and monitor cracks and other defects from non-destructive methods as visual testing (VT) and ultrasound testing (UT) using any known non-destructive testing device.

(54) The integral system and method of the present disclosure is not found in the prior art, and enables one to obtain a suitable internal surface finish of the coke drum by means of water blasting, sandblasting, shot peening, material impacting or laser cleaning as disclosed herein above.

Additionally, embodiments of the present disclosure are heretofore unknown in that they use an inspection crawler **1** (FIG. 5A) or tracking device with combinations of non-destructive inspection methods UT (volumetric) and VT on an interior **51** of a delayed coke drum **16** which is entered and positioned inside this pressure vessel in an independent and autonomous manner guided by remote control operation from the outside of the coke drum. This UT (volumetric) and VT inspection crawler **1** is capable of detecting and characterizing internal cracks (regardless of the depth of the crack and the wall thickness of the coke drum **16**) as well as external cracks. In addition, and as part of the present disclosure, embodiments of the system and method are capable of using other non-destructive evaluation methods such as liquid penetrant test (PT). These features make embodiments of the present disclosure unique in automated non-intrusive inspection systems of coke drums. In addition, embodiments of the present disclosure are unique in that they are configured to detect and characterize cracks in the areas of the cone-skirt junction of conical section **18** which are identified as the AID and the AOD by the American Petroleum Institute 1996 API Coke drum survey (2003).

(55) It is an aspect of the current disclosure that from the results obtained with the cleaning system and method of the present disclosure, that it is possible to obtain welds free of coke incrustations that can be inspected through non-destructive testing methods such as UT, VT, PT or other known techniques.

(56) The methodology used for the inspection with penetrating liquids is the following: I. Cleaning of the wall or internal weld of the coke drum by using the cleaning system. II. Application of the liquid PT disclosed herein above III. Cleaning of the area where the liquid PT was applied by using cleaning device **26** (FIG. 3) or cleaning tool **34** IV. Application of a revealing tint as part of the PT V. Interpretation of results and photographic record. VI. Cleaning up the second Developer Dye by using cleaning device **26** (FIG. 3) or cleaning tool **34**.

(57) From the results obtained with the system and method of the present disclosure (with the use

of ultrasonic inspection), the condition of a coke drum **16** can be estimated for their evaluation of fitness for service and in this way increasing the operational reliability of these pressure vessels. This characteristic makes embodiments of the present disclosure inventive in automated and non-intrusive inspection systems for coke drums.

(58) Now referring to FIG. **13**, there is shown a flowchart of an example coke drum cleaning process **1300**. In some implementations, one or more process blocks of FIG. **13** may be performed by a coke drum cleaning system in accordance with the present disclosure.

(59) As shown in FIG. **13**, process **1300** may include providing a coke drum cleaning system having a vertical member, a gear box fixedly attached to a distal end of the vertical member including a joint, a gear box motor coupled to the gear box, an arm pivotally attached at a proximal end to the joint, a sliding member selectively slidably positioned within the arm, and a cleaning mechanism mounted to a distal end of the sliding member (block **1302**). As also shown in FIG. **13**, process **1300** may include introducing the coke drum cleaning system into an interior of the coke drum (block **1304**). As further shown in FIG. **13**, process **1300** may include pivoting the arm from a vertical position to a horizontal position (block **1306**). As also shown in FIG. **13**, process **1300** may include rotating the arm using the gear box motor about the vertical member in the horizontal position (block **1308**).

(60) Process **1300** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein. Although FIG. **13** shows example blocks of process **1300**, in some implementations, process **1300** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **13**. Additionally, or alternatively, two or more of the blocks of process **1300** may be performed in parallel.

(61) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A coke drum cleaning system comprising: a vertical member: a gear box fixedly attached to a distal end of the vertical member including a joint: a gear box motor coupled to the gear box; an arm pivotally attached at a proximal end to the joint and configured to pivot from a vertical position to a horizontal position; a sliding member selectively slidably positioned within the arm; and a cleaning mechanism mounted to a distal end of the sliding member; and wherein the gear box motor is configured to selectively rotate the arm about the vertical member in the horizontal position; and wherein an inspection system is operable following cleaning of the interior surface using the cleaning mechanism adapted for preparation of a coke drum surface for non-destructive testing.
2. The coke drum cleaning system of claim 1 further comprising: a retractable mechanism motor coupled to the sliding member; and wherein the retractable mechanism motor is configured to selectively position the sliding member between a retracted position and a plurality of extended positions.
3. The coke drum cleaning system of claim 2 further comprising an approach sensor configured to sense a wall of a coke drum and is further configured to position the cleaning mechanism proximate a wall of the coke drum with the arm in the horizontal position.
4. The coke drum cleaning system of claim 3 further comprising: a cleaning pipe system coupled to the vertical member and in fluid communication with the cleaning mechanism.
5. The coke drum cleaning system of claim 4 further comprising: a plurality of nozzles positioned on the cleaning mechanism in fluid communication with the cleaning pipe system.
6. The coke drum cleaning system of claim 5 further comprising: a metal cord coupled to the arm

and to the gear box; a ring slidably mounted to the vertical member wherein the metal cord is slidably positioned within the ring; and wherein the metal cord is configured to be slid vertically to pivot the arm from a vertical position to a horizontal position.

7. The coke drum cleaning system of claim 6 wherein the coke drum includes an upper nozzle positioned in a top of the coke drum and wherein coke drum cleaning system is configured to be introduced into an interior of the coke drum through the upper nozzle with the arm in the vertical position.

8. The coke drum cleaning system of claim 6 wherein the arm is positioned in a cleaning position with the arm in the horizontal position.

9. A method of inspecting a coke drum, the method comprising: providing a coke drum cleaning system comprising: a vertical member; a gear box fixedly attached to a distal end of the vertical member including a joint; a gear box motor coupled to the gear box; an arm pivotally attached at a proximal end to the joint; a sliding member selectively slidably positioned within the arm; and a cleaning mechanism mounted to a distal end of the sliding member; introducing the coke drum cleaning system into an interior of the coke drum; pivoting the arm from a vertical position to a horizontal position; and rotating the arm using the gear box motor about the vertical member in the horizontal position; and performing an automated inspection of a coke drum surface after preparation using a cleaning system.

10. The method of claim 9 further comprising: providing a retractable mechanism motor coupled to the sliding member; and positioning the sliding member between a retracted position and a plurality of extended positions using the retractable mechanism motor.

11. The method of claim 10 further comprising: an approach sensor mounted to the arm; sensing a wall of a coke drum using the approach sensor; and positioning the cleaning mechanism proximate the wall of the coke drum with the arm in the horizontal position.

12. The method of claim 11 further comprising: coupling a cleaning pipe system to the vertical member and in fluid communication with the cleaning mechanism.

13. The method of claim 12 further comprising: coupling a plurality of nozzles positioned on the cleaning mechanism in fluid communication with the cleaning pipe system.

14. The method of claim 13 further comprising: a metal cord coupled to the arm and to the gear box; a ring slidably mounted to the vertical member wherein the metal cord is slidably positioned within the ring; sliding the metal cord vertically; and pivoting the arm from a vertical position to a horizontal position.

15. The method of claim 13 comprising any of water blasting, sandblasting, shot peening, laser cleaning, and material impacting on an inner surface of the coke drum.

16. The method of claim 13, further comprising: providing a coke drum cutting tool; replacing a plurality of cutting nozzles in the coke drum cutting tool with a plurality of cleaning nozzles; and cleaning an inner surface of the coke drum using the plurality of cleaning nozzles.

17. The method of claim 13, further comprising preparing an inner surface of the coke drum for non-destructive inspection.

18. The method of claim 13 further comprising performing liquid penetrant testing using the coke drum cleaning system.
